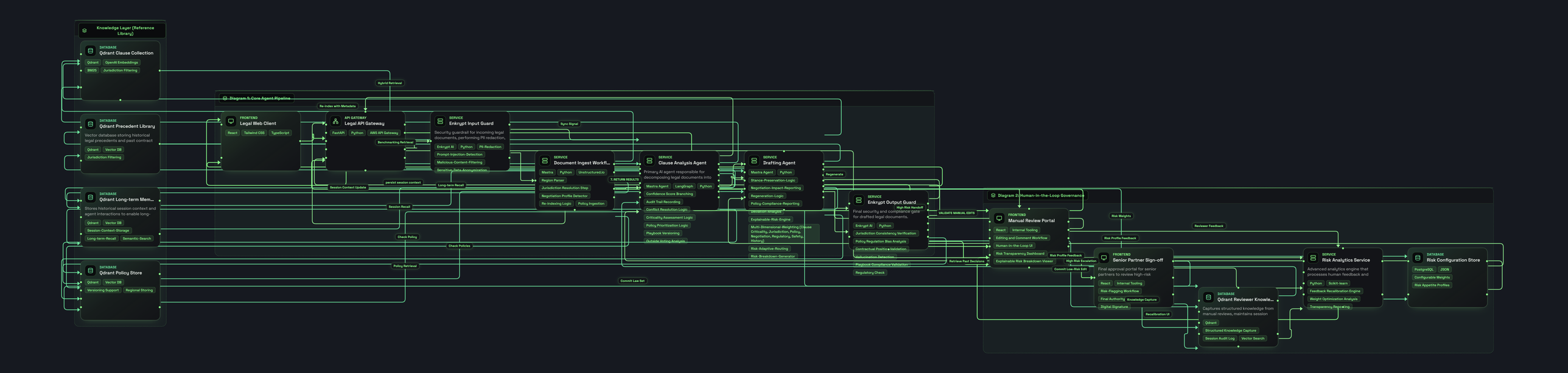
Knowledge Layer (Reference Library)
DATABASE
Qdrant Clause Collection
Qdrant
OpenAI Embeddings
BM25
Jurisdiction Filtering
DATABASE
Qdrant Precedent Library
Vector database storing historical legal precedents and past contract
Qdrant
Vector DB
Jurisdiction Filtering
DATABASE
Qdrant Long-term Mem...
Stores historical session context and agent interactions to enable long-
Qdrant
Vector DB
Session-Context-Storage
Long-term-Recall
Semantic-Search
DATABASE
Qdrant Policy Store
Qdrant
Vector DB
Versioning Support
Regional Storing
Hybrid Retrieval
Diagram 1: Core Agent Pipeline
Re-Index with Metadata
FRONTEND
Legal Web Client
React
Tailwind CSS
TypeScript
API GATEWAY
Legal API Gateway
FastAPI
Python
AWS API Gateway
Benchmarking Retrieval
SERVICE
Enkrypt Input Guard
Security guardrail for incoming legal documents, performing PII redaction,
Enkrypt AI
Python
PII-Redaction
Prompt-Injection-Detection
Malicious-Content-Filtering
Sensitive-Data-Anonymization
Sync Signal
Session Context Update
persist session context
Long-term Recall
Session Recall
Check Policy
Check Policies
Policy Retrieval
SERVICE
Document Ingest Workfl...
Mastra
Python
Unstructured.io
Region Parser
Jurisdiction Resolution Step
Negotiation Profile Detector
Re-indexing Logic
Policy Ingestion
7. RETURN RESULTS
SERVICE
Clause Analysis Agent
Primary AI agent responsible for decomposing legal documents into
Mastra Agent
LangGraph
Python
Confidence Score Branching
Audit Trail Recording
Conflict Resolution Logic
Criticality Assessment Logic
Policy Prioritization Logic
Playbook Versioning
Outside Voting Analysis
SERVICE
Drafting Agent
Mastra Agent
Python
Stance-Preservation-Logic
Negotiation-Impact-Reporting
Regeneration-Logic
Policy-Compliance-Reporting
Jurisdiction-Analysis
Explainable-Risk-Engine
Multi-Dimensional-Weighting (Clause Criticality, Jurisdiction, Policy, Negotiation, Regulatory, Safety, History)
Risk-Adaptive-Routing
Risk-Breakdown-Generator
Regenerate
SERVICE
Enkrypt Output Guard
Final security and compliance gate for drafted legal documents.
Enkrypt AI
Python
Jurisdiction Consistency Verification
Policy Regulation Bias Analysis
Contractual Position Validation
Hallucination Detection
Playbook Compliance Validation
Regulatory Check
High Risk Handoff
VALIDATE MANUAL EDITS
Retrieve Past Decisions
Commit Law Set
Diagram 2: Human-in-the-Loop Governance
FRONTEND
Manual Review Portal
React
Internal Tooling
Editing and Comment Workflow
Human-in-the-Loop UI
Risk Transparency Dashboard
Explainable Risk Breakdown Viewer
Risk Profile Feedback
High Risk Escalation
Commit Low-Risk Edit
Risk Weights
Risk Profile Feedback
Reviewer Feedback
FRONTEND
Senior Partner Sign-off
Final approval portal for senior partners to review high-risk
React
Internal Tooling
Risk-Flagging Workflow
Final Authority
Knowledge Capture
Digital Signature
Recalibration UI
DATABASE
Qdrant Reviewer Knowle...
Captures structured knowledge from manual reviews, maintains session
Qdrant
Structured Knowledge Capture
Session Audit Log
Vector Search
SERVICE
Risk Analytics Service
Advanced analytics engine that processes human feedback and
Python
Scikit-learn
Feedback Recalibration Engine
Weight Optimization Analysis
Transparency Reporting
DATABASE
Risk Configuration Store
PostgreSQL
JSON
Configurable Weights
Risk Appetite Profiles